

HUMAN ANATOMY AND PHYSIOLOGY-II

Instructions to the candidates:

1. Do not write anything on question paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules.
2. Section-A contains Q-1 and Q-2 with Five (5) questions each. All the Ten very short type questions are compulsory and carry equal marks ($10 \times 2 = 20$).
3. Section-B contains Q-3, Q-4 and Q-5. Answer any TWO questions among the three descriptive type questions and carry 10 marks each. ($2 \times 10 = 20$).
4. Section-C contains Q-6 and Q-7 with FIVE (5) and FOUR (4) sub divisions. Answer any FOUR (4) and THREE (3) questions of choice among the short answer type questions and all carry 5 marks each. ($7 \times 5 = 35$)

Duration: 3 Hours

No. of Pages in this Question Paper: 1

Total Marks: 75

Section – A

1. Answer the following questions in very short (Maximum 20 words):

5 x 2 = 10

a) Define Electrophysiology.

b) What is Cushing's syndrome?

c) What is Action Potential?

d) What is vital capacity?

e) What are the functions of cerebrum?

2. Answer the following questions in very short (Maximum 20 words):

5 x 2 = 10

a) What is the role of Insulin and Glucagon?

b) Write various functions of Liver.

c) Write various functions of Cerebellum.

d) What is Oogenesis?

e) What is Micturition reflex?

Section – B

Answer (any Two) of following questions in Long answer:

2 x 10 = 20

3. Draw a well labelled diagram of Brain and explain its structure and functions.
4. Explain anatomy and physiology of Digestive system.
5. Explain the structure of urinary tract with the help of diagram and also write physiology of urine formation.

Section – C

6. Answer (any Four) of the following questions in short answer:

4 x 5 = 20

a) Write the functions of hormones secreted by pituitary gland.

b) Elaborate the role of RAS in kidney.

c) Explain physiology of menstruation.

d) Explain structure and functions of spinal cord.

e) Write the role of ATP.

7. Answer (any Three) of the following questions in short answer:

3 x 5 = 15

a) Explain protein synthesis.

b) Describe various factors affecting respiration.

c) Explain Reflex activity.

d) Classify and write the properties of Nerve fibre.

PHARMACEUTICAL ORGANIC CHEMISTRY-I

Instructions to the candidates:

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3. Section-B contains Q-3, Q-4 and Q-5. Answer any TWO questions among the three descriptive type questions and carry 10 marks each. ($2 \times 10 = 20$).
4. Section-C contains Q-6 and Q-7 with FIVE (5) and FOUR (4) sub divisions. Answer any FOUR (4) and THREE (3) questions of choice among the short answer type questions and all carry 5 marks each. ($7 \times 5 = 35$)

Duration: 3 Hours

No. of Pages in this Question Paper: 1

Total Marks: 75

Section - A

1. Answer the following questions in very short (Maximum 20 words): $5 \times 2 = 10$
 - a) Metamerism
 - b) Diels Alder reaction
 - c) Structure and use of iodoform
 - d) Describe any one qualitative test for aromatic Aldehyde
 - e) Inductive effect
2. Answer the following questions in very short (Maximum 20 words): $5 \times 2 = 10$
 - a) Basicity of amines
 - b) Structure and uses of cinnamaldehyde
 - c) Victor Meyers test
 - d) Ozonolysis
 - e) IUPAC nomenclature of alkanes

Section - B

Answer (any Two) of following questions in Long answer:

 $2 \times 10 = 20$

3. Discuss in detail about E1 reaction. Add a note on factors affecting E1 reaction.
4. Discuss in detail about SN2 reaction. Add a note on SN1 Vs SN2 reactions.
5. Discuss the general methods of preparations of carboxylic acids.

Section - C

6. Answer (any Four) of the following questions in short answer: $4 \times 5 = 20$
 - a) Classification of organic compounds.
 - b) sp^3 hybridization.
 - c) Discuss the general reactions of alcohols.
 - d) Perkin condensation.
 - e) Qualitative tests for aliphatic amines.
7. Answer (any Three) of the following questions in short answer: $3 \times 5 = 15$
 - a) Give the structure and uses of amphetamine, acetyl salicylic acid and benzyl benzoate.
 - b) Cannizzaro reaction.
 - c) Discuss the general methods of preparation of alkyl halides.
 - d) Markownikoff's addition.

BIOCHEMISTRY

Instructions to the candidates:

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4. Section-C contains Q-6 and Q-7 with FIVE (5) and FOUR (4) sub divisions. Answer any FOUR (4) and THREE (3) questions of choice among the short answer type questions and all carry 5 marks each. ($7 \times 5 = 35$)

Duration: 3 Hours

No. of Pages in this Question Paper: 1

Total Marks: 75

Section – A

1. Answer the following questions in very short (Maximum 20 words):

5 x 2 = 10

- | | |
|--------------------------------------|--------------------|
| a) Biological significance of ATP | b) Gluconeogenesis |
| c) Disorders of Vitamin D deficiency | d) Gout disease |
| e) Coenzyme | |

2. Answer the following questions in very short (Maximum 20 words):

5 x 2 = 10

- | | |
|--|--------------------------|
| a) Give an example of competitive inhibition of enzyme | |
| b) Translation | c) Jaundice |
| d) Symptoms of G6PD deficiency | e) Essential amino acids |

Section – B

Answer (any Two) of following questions in Long answer:

2 x 10 = 20

3. Define and classify carbohydrates with suitable examples. Add a note structure of DNA.
4. Discuss the pathway, energetic and significance of citric acid cycle.
5. Discuss in detail about Urea cycle and its disorders.

Section – C

6. Answer (any Four) of the following questions in short answer:

4 x 5 = 20

- | | |
|---|-------------------------------------|
| a) Energy rich compounds | b) Glycogenolysis |
| c) De Novo synthesis of palmitic acid | d) Organization of mammalian genome |
| e) Nomenclature and IUB classification of enzymes | |

7. Answer (any Three) of the following questions in short answer:

3 x 5 = 15

- a) Classification of lipids with suitable examples.
- b) Oxidative phosphorylation and its mechanism.
- c) Protein synthesis inhibitors.
- d) Enzyme kinetics.

28-07-2018

Bachelor of Pharmacy II Semester- End Semester Examination- July - 2018
3624
PATHOPHYSIOLOGY

Instructions to the candidates:

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3. Section-B contains Q-3, Q-4 and Q-5. Answer any TWO questions among the three descriptive type questions and carry 10 marks each. (2 x 10=20).
4. Section-C contains Q-6 and Q-7 with FIVE (5) and FOUR (4) sub divisions. Answer any FOUR (4) and THREE (3) questions of choice among the short answer type questions and all carry 5 marks each. (7 x 5 = 35)

Duration: 3 Hours

No. of Pages in this Question Paper: 1

Total Marks: 75

Section – A

1. Answer the following questions in very short (Maximum 20 words): 5 x 2 = 10
 - a) The most common features of epilepsy.
 - b) Define Hypertrophy.
 - c) How is Parkinson disease treated?
 - d) What is Atherosclerosis?
 - e) What is the main complication in diabetes?
2. Answer the following questions in very short (Maximum 20 words): 5 x 2 = 10
 - a) Write the symptoms of apoptosis.
 - b) Define Metaplasia.
 - c) Write the name of female sex hormones.
 - d) What is Angina?
 - e) What is sickle cell anaemia?

Section – B

Answer (any Two) of following questions in Long answer:

2 x 10 = 20

3. Define cell injury. Give its etiology.
4. Classify cancer. Explain its etiology & pathogenesis.
5. What is AIDS? Write its causative agent, mode of transmission, symptoms, diagnosis & pathogenesis.

Section – C

6. Answer (any Four) of the following questions in short answer: 4 x 5 = 20
 - a) What is Inflammation?
 - b) Give pathophysiology of Peptic ulcer.
 - c) What is Hypertension?
 - d) Write a note on tuberculosis.
 - e) Give Pathophysiology of Rheumatoid Arthritis.
7. Answer (any Three) of the following questions in short answer: 3 x 5 = 15
 - a) Explain Meningitis.
 - b) Explain different type of Hepatitis.
 - c) What is epilepsy? Give their Pathophysiology.
 - d) Give Pathophysiology of Asthma.

Rajasthan University of Health Sciences, Jaipur
Bachelor of Pharmacy II Semester – End Semester Examination – July. 2018

206T

ENVIRONMENTAL SCIENCES

Instructions to the candidates:

1. Section-A contain 2 Questions in which Answer any 1. Each question carries equal marks. $(1 \times 10) = 10$
2. Section-B contain 7 Questions in which Answer any 5. Each question carries equal marks. $(5 \times 5) = 25$

Time: 1:30 hrs

M.M: 35

SECTION A

1. Explain the importance of value education, awareness and community-participation in Environmental protection activities in India. (10).
2. What are various renewal energy options. Explain why their use is important. (10)

SECTION B

1. What is acid rain. What are the causes and effects of acid rain ? (05)
2. What is meant by Bio fuel? How is it important in Environmental conservation? (05)
3. What is the cause of noise pollution? How is noise pollution measured? (05)
4. What are major water pollutants arising out of agriculture practices? (05)
5. Define ecosystem. List the Components of ecosystem. (05)
6. Discuss the relationship between human health and environment. (05)
7. Differentiate between composting and Vermicomposting? (05)

Rajasthan University of Health Sciences, Jaipur
Bachelor of Pharmacy II Semester - End Semester Examination - July, 2018

205T

COMPUTER APPLICATION

Instructions to the candidates:

1. Section-A contain 3 Questions in which Answer any 1. Each question carries equal marks. $(1 \times 10) = 10$
2. Section-B contain 8 Questions in which Answer any 5. Each question carries equal marks. $(5 \times 5) = 25$

Time: 1:30 hrs

M.M: 35

Section-A

1. Explain the following:- . $(2 \times 5 = 10)$
 - i). Convert $(74)_{10}$ to $(?)_8$
 - ii). Convert $(101110010)_2$ to $(?)_8$
 - iii). Difference between octal number system and decimal number system.
 - iv). Convert $(30071)_8$ to $(?)_{10}$
 - v). What is the use of HTML ? Explain
2. Explain XML translation technologies. Design a framework for transforming the structuring of an XML document. (10)
3. What is the use of computer in pharmacy? And also explain the impact of Bioinformatics in Vaccine discovery. (10)

Section-B

1. What is World Wide Web (w3c)? Explain. (05)
2. What is CSS? What are important benefits of using CSS? (05)
3. What is laboratory information management system? Explain. (05)
4. Difference between CDS and TIMS. (05)
5. Explain the concept of information gathering. (05)
6. Explain the process of website development? (05)
7. What is the use of barcode machine? Explain. (05)
8. What is data flow diagram? Explain. (05)